

# UK HEAT SHIFT

## Data sources and estimation methodology

Causeway Energies · [causewaygt.github.io/uk-heatsplit](https://causewaygt.github.io/uk-heatsplit) · **v8.0.0** · August 2026 · [contact@causewaygt.com](mailto:contact@causewaygt.com)

*Supersedes v7.1.0. Changes in this issue: the coincidence premium is withdrawn (former Section 10.7); the calibration ratio now sits outside the publication gate and Section 3 states why; new Section 11 documents the capital panel, including its deployment period and cost-improvement treatment (Section 11.5.1–11.5.2); new Section 12 documents the frontispiece; Section 13–14 renumbered.*

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## Abstract

UK Heat Shift sets out the strategic case for geothermal heating and cooling at gigawatt scale in the United Kingdom, and publishes the estimator on which that case rests. It is a public, automatically updated estimate of the weekly heating and cooling energy balance of Great Britain. Daily gas throughput to the low-pressure distribution network is decomposed into a temperature-sensitive space-heating component and a weather-independent baseline by segmented linear regression on population-weighted degree days, following the established gas-to-heat method of Watson et al. [1,2] and Sansom [3]. The trailing twelve-month space-heating estimate is calibrated against the Department for Energy Security and Net Zero (DESNZ) Energy Consumption in the UK (ECUK) end-use tables [4]. Non-gas fuels and space cooling, which lack daily operational feeds, are held at annual ECUK levels and modulated by degree days. Delivered service is derived from purchased energy using in-situ combustion efficiencies [5] and measured seasonal heat-pump performance factors [6,7]. Cooling demand in the unequipped building stock is estimated independently from overheating degree-hours above the CIBSE comfort threshold [8] and dwelling-stock statistics [9,10]. All emissions use DESNZ conversion factors [11]; all monetary values use the prevailing Ofgem price cap [12]. The headline estimates are additionally published as a weekly historical series over the window the live gas feed serves, priced and carbonated at the rates in force each week. An hourly retrospective tests the transition against the electricity system Britain actually operated, and a capital layer tests whether the resulting energy-bill saving services the investment the transition requires. Every figure resting on author judgement is flagged; the complete source code and data are public. This document specifies each estimator, its inputs, its assumptions, and its uncertainty.

**On advocacy.** This site argues a position: that geothermal heating and cooling deserves a materially larger role in the United Kingdom than it now has. It does so from delivered heat and cooling, measured weekly from live operational data, rather than from resource potential or generation prospectus. Declaring the position raises rather than lowers the standard applied to the arithmetic; the working is published so the conclusion can be contested.

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## 1. Conventions

### 1.1 Sourced quantities and author estimates

Two classes of number appear on the dashboard. Sourced quantities are traceable to a cited publication or a live public feed. Author estimates (marked † on the site) rest on Causeway Energies judgement where no authoritative figure exists; each is accompanied by a stated range and an invitation to challenge. The distinction is maintained throughout this document.

### 1.2 Purchased energy and delivered service

Two physical quantities are tracked and must not be conflated. Purchased energy is metered fuel or electricity at the point of supply. Delivered service is the useful heat or cooling supplied to the conditioned space after conversion. For a combustion appliance of efficiency  $\eta < 1$ , delivered service is less than purchased energy; for a heat pump of coefficient of performance  $COP > 1$ , delivered service exceeds purchased electricity because ambient or ground heat, never purchased, is harvested. The indigenous-supply and efficiency arguments on the site depend on this distinction.

### 1.3 Live estimates and calculators

From v8 the site carries one panel that is not an estimate: the capital panel of Section 11 is a calculator with user-set inputs, and is labelled as such on the page. Everything else regenerates daily from public feeds. The distinction is stated wherever it could be mistaken.

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## 2. Primary heating estimate: gas-to-heat regression

### 2.1 Data source

The primary feed is the National Gas Transmission operational-data REST API [13], from which the daily aggregate demand of the thirteen Local Distribution Zones (LDZ) is retrieved and summed. Publication identifiers are resolved from the API catalogue at runtime rather than hard-coded, so that a renamed publication produces a diagnostic rather than a silent error. The total National Transmission System (NTS) demand is retrieved separately for context. The LDZ aggregate is used in preference to the NTS total because the latter includes gas-fired power generation and directly connected industry, which are unrelated to building heat and would dominate the signal.

### 2.2 Quantifying the LDZ / NTS distinction

The distinction is material. In the current week the summed LDZ demand is of order 480 GWh/day at summer minimum, against an NTS total of order 1,000–1,400 GWh/day; the difference is principally power-station and industrial offtake carried on the transmission system, which does not track heating degree days. Using the NTS total would introduce a large, weather-insensitive term and depress the apparent temperature sensitivity of demand.

## 2.3 Temperature index

Outdoor temperature is represented by a population-weighted mean over twelve Great British urban centres (London, Birmingham, Manchester, Leeds, Glasgow, Newcastle, Bristol, Cardiff, Edinburgh, Southampton, Nottingham, Sheffield), with weights from Office for National Statistics mid-year population estimates. Hourly and daily 2 m air temperatures are drawn from the ERA5 reanalysis [14] via the Open-Meteo archive API [15]. Heating degree days at base temperature  $T_e$  are computed as  $HDD = \max(0, T_e - \bar{T})$ , with  $\bar{T}$  the weighted daily mean; cooling degree days use  $CDD = \max(0, \bar{T} - T_{c})$ . Bases are evaluated at 14.5, 15.5 and 16.5 °C for heating and 18 °C for cooling; the heating base is selected on goodness of fit.

## 2.4 Estimator

Daily LDZ demand  $G$  is modelled as a segmented (piecewise-linear) function of temperature, equivalent to an ordinary-least-squares regression on the degree-day transform:

$$G_i = \alpha + \beta \cdot HDD_i + \varepsilon_i$$

where  $\alpha$  (GWh/day) is the weather-independent baseline attributed to water heating, cooking and other non-space-heating gas use;  $\beta$  (GWh per degree day) is the marginal space-heating response; and  $\varepsilon$  is a residual. The fitted space-heating component on day  $i$  is  $\beta \cdot HDD_i$ , and the baseline is  $\alpha$ . This is the decomposition of Watson et al. [1,2], itself building on Sansom [3]; the method is standard in the GB heat-demand literature. Current fitted parameters are  $\alpha \approx 476$  GWh/day,  $\beta \approx 130$  GWh/HDD, with base  $T_e = 16.5$  °C over a 365-day window (Figure 1).

## 2.5 On the coefficient of determination

The regression returns  $R^2 \approx 0.92$ . Two caveats attach. First, because HDD and demand share a strong annual cycle, a high  $R^2$  is expected and does not by itself validate the functional form. Second,  $R^2$  is inflated by the seasonal range; the residual standard error (of order 180 GWh/day) is the more informative figure for daily uncertainty and is reported in the calibration diagnostics.

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## 3. Calibration to national statistics

The regression yields a self-consistent daily signal but not an absolute national total. The trailing twelve-month integral of the fitted space-heating component is therefore reconciled against ECUK [4], which reports domestic and non-domestic energy consumption by end use and fuel, derived by DESNZ from the Digest of UK Energy Statistics (DUKES) national energy balance combined with the Cambridge Housing Model and the National Energy Efficiency Data-Framework. ECUK space-heating totals are themselves modelled estimates, weather-corrected to a long-run average using the same degree-day apparatus, not metered heat.

The calibration ratio  $r = (\text{modelled 12-month space heat}) / (\text{ECUK anchor, GB-adjusted and weather-normalised})$  is monitored continuously. The publication gate is  $|r - 1| \leq 0.10$ .

### 3.1 The anchor and its weather normalisation

The anchor is ECUK 2025, tables U3 and U5, calendar year 2024, United Kingdom. It is multiplied by a GB share to remove Northern Ireland, then weather-normalised by the ratio of trailing-twelve-month heating degree days to anchor-year heating degree days — asking what the ECUK reference year would have consumed in the weather just experienced.

### 3.2 The gate is currently failing, and why

**The current value is  $r \approx 1.10$ , marginally outside the  $\pm 10\%$  gate.** The live model runs about ten per cent above the ECUK anchor. This is reported openly on the site — the page states that the model is outside its own publication gate and that the split is under review — rather than silently rescaled.

The step across the gate came from a correction rather than a divergence, and the sequence is documented because it bears on how the figure should be read. The degree-day series used to compute anchor-year HDD was fetched over a fixed trailing window. As real time advanced, that window rolled past 1 January of the anchor year, and the anchor-year degree-day total was progressively truncated. A truncated denominator inflates the weather-normalised anchor and flatters the ratio downward. Computing the fetch span from the anchor year itself restored the full year: anchor-year HDD rose, the normalised anchor fell, and the ratio stepped from approximately 1.05 to approximately 1.10. A guard now refuses to normalise on an incomplete anchor year, warning instead.

The correct reading is therefore that the model has been running approximately ten per cent above ECUK for some time, and that the truncation was concealing it. The figure did not deteriorate; a measurement error was removed.

### 3.3 Candidate contributions to the residual

Stated as candidates, not findings.

The regression attributes **all** HDD-correlated LDZ gas to space heating. Hot-water demand genuinely rises in cold weather as incoming mains water cools, and any weather-correlated commercial or process load enters the same slope. This biases the model upward and is the first hypothesis under test.

A linear fit through a year including summer zeros can overstate the slope where the underlying relationship has curvature near the base temperature; the residual standard error of order 180 GWh/day is not small.

The base temperature is selected on fit each run, and both trailing-window and anchor-year degree days are computed at whichever base wins. If the winning base moves, the two totals change but not proportionally, and the ratio steps. This sensitivity is not currently logged and should be.

Working against all of the above, the anchor year precedes the trailing window by roughly two years while underlying gas heat demand is falling, which should depress the ratio. The residual exists despite that.

The anchor carries a domestic/services split from the same U3/U5 tables. If the services leg is the part that does not reconcile — consistent with the first hypothesis — that split is where the resolution lies.

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## 4. Cooling: the equipped fleet (observed)

### 4.1 Data source and reconstruction

Cooling delivered by installed plant is inferred from national electricity demand. Half-hourly demand is taken from the National Energy System Operator (NESO) historic demand data and rolling update files [16]. Metered demand understates true underlying demand by the quantity supplied behind the meter, principally embedded solar photovoltaics, which is largest precisely on the hot, bright days when cooling load peaks. Underlying demand is therefore reconstructed by adding NESO estimates of embedded solar and wind generation to metered demand before analysis. The correction is material: embedded solar alone reaches of order 10–16 GW at midday in high summer against a metered demand of order 20–30 GW, so the hot, bright hours in which cooling peaks are precisely those the meter understates most. Future-dated rows present in the update file, and days with other than 48 half-hourly values, are removed.

### 4.2 Within-class centring

Summer electricity demand carries a strong non-cooling structure: it falls during the August holiday period and differs between weekdays and weekends. Binning raw demand against CDD conflates this structure with the cooling response and can yield a spurious negative uplift. To isolate the cooling signal, each daily underlying-demand value  $d_i$  is expressed as an anomaly about the mean of its (calendar-month  $\times$  weekend/weekday) class:

$$a_i = d_i - \bar{d}(m_i, w_i)$$

where  $\bar{d}(m, w)$  is the mean underlying demand over all days sharing month  $m$  and weekend/weekday status  $w$ . This within-class centring removes the seasonal and holiday baseline while preserving the temperature-driven excursion. The operation is the within transformation of standard panel econometrics — a two-way fixed-effects demeaning on calendar month and weekend/weekday class [31] — applied here to isolate a weather signal rather than to estimate a panel coefficient: the estimator is textbook, its application to GB cooling demand is this site's own, and it is validated against synthetic data with a known injected slope in Section 4.4. The term “demeaning” is avoided in public-facing copy in favour of “within-class centring” to prevent confusion with the ordinary meaning of the word.

### 4.3 Response curve and latent slope

The anomalies  $a_i$  are grouped into CDD bins; the mean anomaly per bin is the observed cooling response. A linear slope fitted to the low-CDD bins, where the fleet operates unconstrained, gives the latent response in GWh per CDD; extrapolating it defines the demand that the equipped fleet would meet in the absence of capacity or behavioural limits. Departure of the observed curve below this line at high CDD would indicate saturation. The current curve does not saturate — the highest bin shows the largest uplift — consistent with field evidence that

the installed UK cooling stock is not capacity-constrained at present temperatures, and implying headroom rather than a ceiling.

#### 4.4 Validation

The centring estimator was validated before deployment on synthetic data in which a known cooling slope was combined with an injected holiday depression of comparable magnitude to that seen in real August demand. Naive binning of the synthetic series reproduced the spurious negative-uplift artefact; within-class centring recovered the injected slope to within sampling error. This confirms that the procedure removes the confound it is designed to remove without distorting the underlying signal.

#### 4.5 Relationship to the billed cooling figure

The observed electricity response yields a weekly cooling electricity of order 150–230 GWh, whereas the ECUK-anchored figure used in the cost and carbon panels is of order 1,000–1,200 GWh. This order-of-magnitude discrepancy is stated on the site and left unreconciled pending a full cooling season. Both estimates may carry error. The observed figure captures only the marginal, weather-driven increment in national demand and will understate cooling to the extent that base-load cooling (data centres, process refrigeration, continuously conditioned commercial space) is weather-insensitive and absorbed into the class means. The ECUK-shaped figure, conversely, distributes an annual national cooling estimate across the year by CDD and may overstate a given summer week if the annual ownership or intensity assumptions are high, or if the CDD base temperature (18 °C) misrepresents the temperature at which conditioning actually begins.

From v5.2 the planned resolution runs as a published diagnostic: a year-round ordinary-least-squares fit of daily underlying demand on HDD and CDD jointly — the CDD base swept from 10 to 18 °C and chosen on fit, so the data decides where conditioning starts — with weekend, August-holiday and Christmas-period dummies and a linear trend. The fitted cooling slope multiplied by trailing twelve-month CDD bounds annual weather-driven cooling electricity; the remainder against the ECUK anchor is weather-flat, an upper bound on ventilation plus base-load cooling plus below-base conditioning that this method cannot separate (†). The diagnostic has run continuously since v5.2 and is published today inside the cooling panel’s evidence fold, alongside the summer centring slope, as an internal consistency check. It does not drive the bill or carbon panels. The switchover decision is scheduled for September 2026, once a full observed cooling season (May–September) is in hand; if taken, it would restate the summer cooling figures and the shares derived from them by the same documented restatement mechanism used for the weekly history.

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### 5. Cooling: the comfort deficit (modelled)

The observed method sees only buildings that possess cooling. Since fewer than 5% of UK dwellings are air-conditioned [17], the electricity signal is structurally blind to the majority of the population experiencing summer overheating. This latent demand is therefore estimated independently, and explicitly as a stock-and-engineering calculation rather than an inference from the electricity data.

Overheating exposure is quantified as overheating degree-hours above the CIBSE operative-temperature threshold of 26 °C for occupied spaces [8], population-weighted, from the same hourly ERA5 series:

$$\text{ODH}_{26} = \sum_h \max(0, T_h - 26)$$

summed over hours  $h$  in the week. All hours are counted; an occupied-hours weighting (a candidate refinement, †) would reduce the total. The 26 °C threshold is a comfort-risk criterion, not a thermostat setting: newly installed air conditioning is typically set nearer 22–24 °C and would run for materially more hours than this tier counts. The direction is unambiguous — a lower switch-on temperature raises both hours and load — and the scale is large: relaxing the threshold from 26 to 24 °C roughly doubles the overheating degree-hours of a typical GB summer week (†), with load and cost rising in proportion. This tier is therefore a conservative floor on latent demand rather than a central estimate of what installed equipment would draw. The latent thermal demand is  $\text{ODH}_{26}$  scaled by the affected, unequipped dwelling stock and a per-dwelling thermal-response coefficient, plus an equivalent term for uncooled non-domestic comfort floor area derived from the Building Energy Efficiency Survey [18].

### 5.1 Uncertainty on the stock and response assumptions

The dominant uncertainties are the overheating-prevalence fraction and the thermal-response coefficients, all author estimates (†). Three prevalence scenarios are carried and never averaged: a low case of 11% of dwellings, from English Housing Survey self-reported overheating [9]; a high case of over one-half, from the Climate Change Committee assessment of dwellings at overheating risk [10]; and a central case of 25% (†). The per-dwelling response (order 0.2 kWh per degree-hour †) and the non-domestic response (order 3 Wh m<sup>-2</sup> per degree-hour †, over ~190 Mm<sup>2</sup> †) are engineering judgements with a plausible factor-of-two spread each. These ranges bound the latent-demand estimate; they are independent of, and do not narrow, the observed-versus-ECUK discrepancy of Section 4.5, which concerns the equipped fleet rather than the deficit.

### 5.2 Delivery efficiency and the geocooling comparison

The central latent thermal load is expressed as the electricity required to meet it by two routes: passive or near-passive ground cooling at an effective COP of order 20 (†), and air-source vapour-compression at a seasonal EER of order 3.5. The resulting six-to-one ratio quantifies the electrical advantage of ground-coupled cooling. The seasonal-storage corollary — that heat rejected to an aquifer or borehole field in summer is recoverable in winter at a round-trip thermal efficiency of order 70% (literature range 50–80%) [19] — is shown in the dashboard's seasonal-mirror graphic with the loss represented explicitly.

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## 6. Conversion, cost and carbon

### 6.1 In-situ efficiencies and performance factors

Combustion appliances are derated by in-situ, not nameplate, efficiencies: gas 0.835 and oil ~0.82 [5]. Heat pumps are credited with a seasonal performance factor of 3.24 for ground-



source systems, from the Energy Systems Catapult in-situ monitoring of more than 1,700 installations [6]; the associated cold-weather stability of ground-source relative to air-source is documented in the Electrification of Heat demonstration [7]. These are the values underlying the delivered-service bars and the daily spark gap.

## 6.2 Cost

Retail costs are priced by sector. Each fuel component is split into its domestic and services shares using the same ECUK U3/U5 tables that anchor the mix; the domestic share is priced at the prevailing Ofgem default-tariff cap unit rates [12], refreshed quarterly, and the services share at non-domestic rates anchored on the DESNZ Quarterly Energy Prices non-domestic tables [30] (Causeway estimates †, set within the published size-band range — the Q4 2025 manufacturing averages of 3.5 p/kWh gas and 16.8 p/kWh electricity excluding CCL form the large-user floor, with services buildings sitting in the higher small and medium bands — refreshed annually). Cooling, an ECUK services-only category, is priced wholly at the non-domestic electricity rate. Standing charges are excluded throughout. The daily spark gap uses wholesale commodity prices only — National Gas System Average Price [13] and the Elexon market index [20] — divided by the respective efficiency or performance factor, and excludes network, policy and supply costs. Oil is priced from published heating-oil surveys (†).

The spark gap is volatile day to day: the electricity index has moved by more than twenty per cent between consecutive days, changing the gap by a third with nothing else altering. It is a live figure and should be quoted as one.

## 6.3 Carbon

Combustion emissions use DESNZ (Defra) greenhouse-gas conversion factors [11], reported on a gross calorific value basis consistent with UK metered gas billing: natural gas 0.18296 kgCO<sub>2e</sub>/kWh (2025), kerosene ~0.247, coal ~0.345. Electric routes use the live GB grid carbon intensity from the NESO Carbon Intensity API [21], as a seven-day mean. Bioenergy is counted at zero combustion emissions under the biogenic convention, excluding supply-chain emissions — a stated simplification. Because combustion factors are fixed by fuel chemistry while the electric factor tracks the decarbonising grid, the heat-pump rows fall over time and the combustion rows do not; the panel is designed to make this divergence legible.

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## 7. Indigenous-supply share

The headline indigenous share is computed on a delivered-service basis: each unit of delivered heat or cooling inherits the UK-origin fraction of its energy input — gas ~42% (UK-origin share of the gross supply pool — UKCS production plus grid biomethane over production, imports and stock draw — DUKES 2026 [22]), grid electricity ~75% (†, from generation-mix and interconnector data), harvested ambient and ground heat 100%. The last follows the Eurostat and DUKES convention of treating captured ambient heat as indigenous renewable supply [22,23]. Cooling's delivered multiple inherits the origin share of its input electricity; thermodynamic leverage is not treated as an energy origin. The purchased-energy-basis share is retained in the published data for continuity; it changes little under a geothermal shift



because ground heat is never purchased, which is the substantive point the services basis is designed to expose.

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## 8. Other panels

### 8.1 Non-gas fuels

Oil, bioenergy, solid fuel, heat networks and electric resistive heating lack daily operational feeds and are held at ECUK annual levels [4], modulated where appropriate by degree days. They move slowly and contribute little week-to-week variance.

### 8.2 Geothermal

The present geothermal contribution is anchored on the EREC 2025 UK Country Update [24]: approximately 1.43 TWh/yr from an installed base of order 55,000 ground-source heat-pump units, plus of order 0.07 TWh/yr from deep, mine-water and open-loop district schemes (Gateshead, Eden, and others). The stated range is 1.4–2.0 TWh/yr (†); no national register or metering of ground-source output exists, and the Environment Agency’s operational-unit estimate is lower than the installed-unit figure, so this is the most weakly constrained quantity on the site. Forward bars for 2027 and 2031 are trend and scenario projections (†) informed by the EREC Geothermal Market Report 2025 [25] (4,070 UK unit sales in 2025; four new large closed-loop systems; ~11 district systems in development) and the Climate Change Committee Seventh Carbon Budget pathway [26]; the 2050 bar is the Project InnerSpace/REA/ARUP ambition [27]. The European benchmark (2.55 million units delivering 88 TWh in 2025; Swedish sales of 26,785 units against UK sales of 4,070) is drawn directly from [25].

### 8.3 The empty bar: installed hardware vs the what-if

The 20% what-if is also priced in hardware. Serving one-fifth of delivered buildings heat at 2,000 equivalent full-load hours (†) requires installed thermal capacity of order 30–40 GWth (the exact figure is computed live from the model’s delivered-heat estimate); operating UK ground-source capacity is 861 MWth (55,210 systems, EGC 2025 country update), with deeper geothermal adding of order 10 MWth (†, Gateshead, Eden and peers; the EGC table reports only 1.4). The panel sets these against installed reality in France (2,293 MWth ground source + 724 deep), the Netherlands (2,486 + 367 — the Dutch deep capacity being almost entirely greenhouse heat rather than buildings) and Sweden (8,120 + 47), from the EGC 2025 country update summary (Sanner et al., Tables 3–4, end-2024), constants shared with the Irish sibling site. The per-person framing is the substantive point: Sweden operates roughly 778 installed thermal watts per person against the UK’s ~13 (†), so the what-if is unremarkable by European precedent and large only in aggregate. A companion calibration restates each installed fleet as a share of its own country’s buildings heat (space plus hot water, input basis; national anchors are order-level estimates † — UK live from this model, France ~350 TWh, Netherlands ~115 TWh, Sweden ~80 TWh, per IEA, ODYSSEE and Heat Roadmap Europe sources): the UK sits at ~0.5%, France ~2%, the Netherlands ~5%, and Sweden ~20% — Sweden’s existing fleet already serves approximately the share the UK what-if proposes, which is the panel’s answer to the scale objection. Cooling capacity is excluded throughout.

**A comparison withdrawn in v8.** Earlier drafts set the what-if's installed capacity against the roughly 35–40 GW of combustion capacity the UK installs annually replacing boilers, concluding that the what-if was “about one year of that replacement flow”. The two figures are not on a common basis: the what-if capacity is an energy sizing at 2,000 equivalent full-load hours, while boiler nameplate capacity is set by instantaneous hot-water draw and runs at an implied utilisation nearer 560 hours. On heat delivered, one year of the replacement flow serves of order 20–24 TWh against the what-if's ~74 TWh — about three to three and a half years, not one. The comparison is retained only in the corrected form.

## 8.4 Northern Ireland

Northern Ireland operates separate gas and electricity systems and the all-island Single Electricity Market, and is not covered by the GB live feeds. Its heat is estimated annually from the Department for the Economy energy statistics and NISRA Continuous Household Survey [28], modulated by NI degree days. Oil heats just over 60% of NI dwellings and has no meter or daily feed; the resulting data gap is itself reported. Northern Ireland is covered properly by the companion site, Irish Heat Split, which applies the same estimator family to the island of Ireland on its own feeds and conventions; the two sites share constants where the underlying quantity is shared and diverge where the systems do, and a cross-calibration between them runs as a regression test in the build checks. Cooling is deliberately not compared across the two: the island figure counts the cold economy (data centres, refrigeration, process), the GB figure counts ECUK comfort cooling and ventilation.

## 8.5 Whole-economy context (WHY HEAT?)

Four annual pies place heat beside transport and non-heat electricity across delivered services, spend, imported energy and emissions. Sourced anchors are ECUK 2025 final-energy and transport totals [4] and DESNZ provisional greenhouse-gas statistics by sector [29]; the service-level allocations (heat total ~630 TWh, spend, import and emission splits) are author derivations (†), deliberately rounded. Non-energy emissions are excluded; the view is of energy services, not a national inventory.

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# 9. Historical series (the trend layer)

## 9.1 Construction

From v5.0 the four headline estimates — purchased energy, indigenous share, national bill and emissions — and their 20%-geothermal what-if counterparts are also published as a weekly historical series in the daily data file, rendered on the site as a sparkline under each headline with a selectable one-week to twelve-month window. The series is built from calendar weeks (Monday–Sunday); a week enters only if the gas feed served all seven of its days, so every published point is live by construction — no modelled or interpolated weeks appear. Each point is computed by the same estimator code path that produces the current-week headline figures: one implementation, applied to a different week, rather than a parallel reconstruction that could drift. Entries are frozen as first published, except the two most recent weeks, which are recomputed each day against upstream feed revisions; the series is capped at 60 weekly entries. The stored schema has evolved by documented one-time restatements (a version field records

the current schema): each restatement recomputes every stored week still inside the live gas window through the same estimator path, reusing each week's stored grid intensity and the tariff in force, so no historical inputs are refetched or altered. Restatements to date added one-decimal precision to the indigenous share; per-week heat/cooling splits of purchased energy, bill and emissions for both the actual and what-if series; and, at schema 7, a per-fuel block of purchased and useful energy on every entry, so the energy-balance bars can be re-totalled over any trend window rather than showing one week's shares beneath a twelve-month headline. Weeks the feed window has rolled past are unrecomputable and retain their original form, stated where it affects a displayed total.

## 9.2 Time-consistent pricing and carbon

Two inputs vary materially within the window and are applied at their historical, not current, values. Monetary figures for each week use the Ofgem default-tariff cap unit rates in force during that week, from a maintained table of cap periods, so the bill series reflects price-cap movements rather than repricing the past at today's rates. Emissions for each week use that week's mean GB grid carbon intensity, computed as the mean of the half-hourly NESO actuals for the seven days [21], so the electric-route contribution falls through the window as the grid decarbonises. Estimator constants for which no weekly series exists — non-cap fuel prices (heating oil, solid fuel, bioenergy), the non-domestic gas and electricity rates and sector shares of the bill blend (Section 6.2), the ECUK anchor vintage, and the indigenous input-origin shares — are today's values applied uniformly across the window. This is stated alongside the series; it bounds the interpretation of the bill and indigenous-share histories more than the energy and emissions histories, and it means the earliest weeks of the window are not exact reproductions of what the dashboard published at the time.

## 9.3 The live/modelled boundary

The National Gas operational feed serves approximately the trailing thirteen months of daily LDZ demand. The feed's actual served window is recorded at every build in the published data (oldest and newest dates served, and the day count), so the boundary of the live series is empirical rather than assumed. The trend layer publishes nothing beyond that window. A modelled back-cast — the same regression applied to archived degree days (ERA5 reaches 1940; NESO carbon-intensity archives reach 2018) and calibrated to the ECUK vintages then current — remains a planned second phase and is not built. What v7 delivered instead is depth rather than length: the thirteen-month hourly retrospective of Section 10, built on the same archives, which establishes the fetching, storage, restatement and gating machinery a back-cast would need. Outstanding for the back-cast are the historical ECUK vintages, a per-vintage calibration path, and the distinct visual grammar; live and modelled points will never be blended undistinguished.

## 9.4 Presentation

Each sparkline shows the actual series as a solid line and the 20%-geothermal what-if as a lighter line, with the band between them shaded: the vertical gap is the estimated forgone saving in that week, and its integral over the window is the cumulative case for the counterfactual. The change since the window start is stated per headline — in percentage points for the indigenous share, in per cent otherwise — and coloured by whether the

movement is welcome (a rising indigenous share; falling energy, bill and emissions), not by its sign. The window selector also re-totals the headline figures and their what-if counterparts: sums of purchased energy, bill and emissions over the complete live weeks of the window (four, twelve or fifty-two), the indigenous share as a purchased-energy-weighted average, and the what-if row restated as the cumulative forgone saving; every splittable figure carries its heat/cooling split in each window's caption, summed from the per-week stored splits. From schema 7 the energy-in and energy-out bars re-total over the same window from the per-week per-fuel blocks. At the default this-week view the headline figures are the live estimates and the sparkline shows the full record as context.

**From v8 the section heading names the selected window**, and the selector sits directly beneath it, so a reader arriving at a re-totalled figure can see at a glance what period it covers.

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## 10. The hourly retrospective and the system view

### 10.1 Purpose and construction

The panels of Section 1–9 describe the transition in weekly aggregate. Section 10 tests it against the electricity system Britain actually operated, hour by hour, across the trailing twelve months. A separate engine reconstructs an hourly retrospective and stores it alongside the weekly file: temperature (ERA5, twelve population-weighted points, identical to the degree-day layer), wholesale price (Elecon market index), grid carbon intensity (NESO), observed underlying demand, and observed wind and solar generation. Thirteen months are held so that twelve complete calendar months always exist for the seasonal exhibit; every annual statement is computed on the trailing 8,760 hours.

### 10.2 Hourly shaping of heat demand

Annual useful space heat is distributed to days in proportion to each day's share of heating degree days at the live regression base, then within each day in proportion to heating degree-hours; hot water is a flat hourly rate. Two conventions govern the result. First, days below a threshold of one heating degree day contribute no space heat, with a smooth ramp to full contribution at three (†): raw degree-hour allocation assigns winter-strength heat to cool summer nights, which is behaviourally implausible and inconsistent with the weekly estimator, whose baseline subtraction reports no space heating in those weeks. Second, the weights are normalised after damping, so the trailing year still integrates exactly to the annual anchor — the damping redistributes heat into the heating season rather than discarding it. The resulting shape is modelled, not metered, and is marked as such wherever it is drawn.

### 10.3 Coefficient of performance and calibration

Each replacement route is evaluated hourly as a Carnot fraction:  $COP = \eta \times (T_{\text{flow}} + 273.15) / \max(8, T_{\text{flow}} - T_{\text{source}})$ , with source temperatures of outdoor air less a 3 K approach for air-source, 8 °C for shallow ground and 10 °C for geothermal networks (†). Flow temperature is weather-compensated from 30 °C at +15 °C outdoor to 50 °C at –5 °C for space heating, and fixed at 52 °C for hot water year-round (†); each hour's electricity is the load-weighted sum of the two legs. Source temperatures enter every hourly COP directly through the lift term, not as a

label: air-source follows the outdoor series hour by hour, ground and network sources are held constant at their stated values. Ground at 8 °C is a UK shallow-borehole annual mean; 10 °C for networks reflects an ambient loop rather than a deep resource, and is conservative — a deeper or waste-heat-fed network would run warmer. That conservatism costs little either way, because the seasonal performance anchor pins the annual: raising the network source from 10 to 15 °C lowers the fitted  $\eta$  to compensate and moves the worst-hour addition from 6.8 to 7.1 GW — that is, a warmer assumed source makes the published winter figure very slightly worse, not better, so the choice cannot flatter the case. The flow regime is deliberately the same for all three routes: the what-if is a replacement onto the existing emitter stock, so weather compensation applies whatever the source. A flat performance factor for networks was considered and rejected on that basis, and because it would have improved the figure most favourable to the thesis by roughly a quarter. Air-source additionally carries a defrost band peaking near 2 °C, shape taken from the open monitored fleet at [heatpumpmonitor.org](http://heatpumpmonitor.org) and level re-anchored by calibration (†). The scale factor  $\eta$  is then solved per route by bisection so that the trailing year integrates exactly to the published seasonal performance factors of 2.80 (air-source), 3.24 (ground source) and 5.0 (networks, a site convention †). Because hot water is a harder duty than weather-compensated space heating, the split raises  $\eta$  and lowers the winter route loads relative to a single-flow model; it also reverses the summer ordering, air-source outperforming ground source above roughly 11 °C — the temperature at which outdoor air, less its approach, exceeds the ground temperature.

#### 10.4 Stress, netting and the binding hour

Two hourly statistics are published. The first is the coldest hour: the year's highest modelled heat demand, with each route's replacement electricity expressed against that hour's observed demand. It is a gross figure. The second is the binding hour: for each route, the hour maximising observed demand plus the route's net addition less observed wind and solar, searched across November to March. Net here subtracts the fifth of existing resistive space heating the what-if displaces, shaped like space heat and route-independent; that displaced load is itself space-shaped, and night-storage tariff profiles would reduce the credit at morning and evening peaks (†), so the gross figures bound the zero-credit case. The two statistics need not fall in the same hour, or even the same week, and their ratios differ: the coldest-hour figures are additions alone and separate by the ratio of coefficients of performance, while the binding-hour figures are total system requirements dominated by a common standing demand.

#### 10.5 The electrification limit

From v7.1 the panel reports how far Britain could electrify its building heat before the winter peak fills the existing fleet, per route, expressed both as a share of building heat and in TWh a year. A companion figure repeats the calculation with observed wind replaced by a no-wind-all-winter counterfactual, and by intermediate caps, so the reader can see how much of the headroom is weather rather than plant.

Three framings must travel with the figure and are printed beside it. It is a **peak-capacity test, not an energy test**: electrifying all building heat through networks would still require of order 80 TWh a year of additional generation, which this statistic does not address. Distribution networks are unmodelled, and are likely to bind earlier for routes that place plant behind millions of individual low-voltage connections. And one winter is one sample.

## 10.6 The capacity line

Requirements are set against a capacity line constructed as a fixed de-rated block of dispatchable and interconnector capacity plus the wind and solar actually generated in each hour. The block is taken as 62 GW (†) from the NESO Winter Outlook 2025/26: a base-case de-rated margin of 6.1 GW at 10.0% of average cold spell peak implies peak demand of 61.0 GW and de-rated supply of 67.1 GW, from which the Outlook's own wind and solar de-rating of approximately 5.1 GW is removed to avoid double counting. Taking renewables as observed rather than de-rated is deliberate: it removes the most contested assumption in margin arithmetic, and the line then sags exactly when the wind did not blow. **The block is a constant and must be revised each autumn** against the incoming Winter Outlook; it is winter-assessed, so all capacity statements are confined to November through March.

## 10.7 What the capacity comparison is not

It is a static overlay, not a dispatch model. No redispatch, interconnector response, demand-side response or price response is represented; distribution-level constraints are unmodelled; and a capacity expressed in gigawatts cannot represent storage duration — batteries can serve a peak hour but not a fifth consecutive cold day, which is what the exhibit week shows. Exceedance is therefore reported as hours in which a route's requirement exceeds available headroom under today's fleet, never as any statement about supply interruption. One winter is one sample: in 2025-26 the cold snap was windy and the still spell (18–21 March, wind falling to 1.6 GW) was mild, so the coincidence of the two remains a risk the reader may construct but this record does not demonstrate.

## 10.8 Verification

Four gates run before any hourly output is published. G1 confirms each route's calibrated performance factor reproduces its anchor. G2 reconciles the hourly network-route electricity against the weekly estimator's equivalent figure; it is near-tautological by construction and therefore proves wiring and heat conservation rather than independent physics. G3 confirms cooling electricity is conserved against its annual anchor. A feed gate refuses any series missing more than five per cent of its hours. Failure of any gate leaves the previous hourly output in place and the weekly panels untouched. Independent corroboration comes from elsewhere: the located worst hour falls inside the peak week that the weekly gas estimator identifies separately, and on the same day as the year's record observed demand hour.

## 10.9 A statistic withdrawn: the coincidence premium

From v7.0 to v8.0 the panel published a coincidence premium: each route's replacement electricity priced at the wholesale index of the hour in which it was drawn, against the same energy at the flat mean price — the cost of *when* heating draws power rather than how much.

Computed over the trailing year, the statistic proved unstable and eventually self-contradictory. Heat is drawn overwhelmingly in winter, but the annual mean price against which it was compared included summer hours in which almost no heat is drawn; as summer scarcity pricing rose, the annual comparator rose with it and the premium fell, crossing zero. A negative coincidence *premium* is a contradiction on the page.



Restricting the calculation to the heating season (October–March, matching the Winter Outlook window that sets the capacity line) resolved the sign: priced against the mean of the same months, the premium was small and positive for all three routes, and largest for air-source, as the mechanism predicts. This narrows the statistic deliberately — it then measures timing *within* the heating season and excludes the winter-versus-summer price level, which is a seasonal effect rather than a coincidence.

The statistic was nevertheless withdrawn from the page in v8.0: it is difficult to read at a glance and was not carrying weight in the argument. It continues to run on the winter basis as an internal diagnostic and is logged with every build, so the question remains answered and the panel can be restored without repeating the work. This paragraph records the episode because the site’s claim to be checkable rests on documenting corrections, not only results.

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## 11. The capital layer (v8)

### 11.1 What it is, and what it is not

Section 10 asks whether the transition fits the electricity system. Section 11 asks whether it pays for itself. The capital panel answers one question — does the avoided energy cost service the capital the transition requires? — and it is **a calculator, not an estimate**. It is the only panel on the site driven by user inputs, and it is labelled as such on the page. Every other figure regenerates daily from public feeds; this one regenerates from public feeds *and* a set of stated assumptions the reader may change.

The design intent is adversarial in the reader’s favour. The honest answer is a range whose width is dominated by assumptions no feed can settle. A single number would be a false claim of precision; exposing the levers invites the reader to test the conclusion rather than accept it.

### 11.2 The live input

The saving is the difference between what Britain paid for heat and cooling and what the 20% what-if would have paid, summed over the complete live weeks of the trailing twelve months from the same stored history as Section 9. It is **pinned to twelve months regardless of the trend selector**: in an August week the saving is near zero, and a capital figure driven from one such week would swing with the weather rather than the case.

Because the what-if bill is computed on the post-conversion energy mix, the saving is already net of the replacement electricity. It is nevertheless a price differential rather than a physical quantity, and it narrows if the gas–electricity spread narrows.

### 11.3 Unit capital: the basis, and why it is not per kilowatt

Unit capital is held per class in **pounds per annual megawatt-hour delivered**, not per kilowatt of installed capacity. Equivalent full-load hours differ by a factor of three across the relevant classes — of order 1,800 h for a domestic ground-source system against 5,400–6,000 h for a baseload doublet — so a capacity-weighted per-kilowatt blend over-weights individual ground-source by three to four times relative to the heat it actually delivers. Blending on heat

delivered normalises this automatically. Per-kilowatt figures are never blended across classes without their load factors stated.

Class anchors, real 2025 GBP, all daggered:

| Class                        | £ per annual MWh | £/kW        | EFLH          | Evidence                                                                                           |
|------------------------------|------------------|-------------|---------------|----------------------------------------------------------------------------------------------------|
| ATES, above 2 MW             | 100–260          | ~255        | 2,000–2,500 h | Strongest: 133 systems, NL/BE/DK, converging above 2 MW                                            |
| Intermediate doublet (<2 km) | 290–320          | ~1,600      | 5,400–6,000 h | Thinnest: genuine sub-2 km European doublets are scarce; leans on Paris Dogger and Dutch analogues |
| Individual GSHP, vertical    | 700–1,670        | 1,600–3,500 | 1,800 h       | Arup/DESNZ medium construction, corroborated by UK market quotes                                   |

Borehole thermal storage is carried on a per-cubic-metre and per-kilowatt-hour-stored basis, never per kilowatt, and does not enter the blend.

The UK anchor is Arup for DESNZ [32]. European comparators are used where the UK evidence is thin, with the documented UK subsurface premium on intermediate-depth drilling retained inside the doublet range rather than applied separately.

#### 11.4 Distribution, connections and development

**Every published class anchor excludes the pipe.** The Arup construction figure excludes the network; the Dutch subsidy reference installations exclude distribution, building connections, back-up, development expenditure and permits; the ATES figure covers heat pump and heat exchanger only. Omitting distribution is therefore the single largest error available in this arithmetic, and it flatters the case.

Distribution and building connections are carried as **two separate levers**, because they scale on different physical drivers and because the variance lives almost entirely in connections: of order £40 per annual MWh for bulk supply to an anchor load against £900 or more where every dwelling is connected individually. Central values are drawn from UK outturn — DECC/AECOM [33], seven schemes at 2013/14 prices, escalated to 2025 by the RICS BCIS General Building Cost Index — and cross-checked against scheme-level generation-versus-pipework

splits in the DESNZ heat network pipeline [34], which range from about 10% distribution for a dense scheme taking waste heat to about 82% for a small extension serving dispersed housing. **Linear heat density, not technology, is what moves this figure.**

Both adders fall on the **network share only**. An individual ground-source installation has no distribution main and no network connection; its pipework already sits inside its own class anchor, and charging it again would double-count.

Development and permits are added at 20% (†) as a constant rather than a lever, since every class anchor excludes them and the long-standing UK planning heuristic places development at up to a quarter of scheme cost.

Two intuitions are explicitly corrected. An ambient or fifth-generation loop is **not** reliably cheaper per metre than conventional district heating: the small flow-return temperature difference forces larger diameters, and trench excavation dominates cost either way. The genuine ambient advantage is operational and in the long life of the groundworks, not in cheaper pipe. And the denominator is heat **and cooling** delivered: one bidirectional network carries both, so its pipework is not charged to heat alone.

## 11.5 Finance

The saving is capitalised at a user-set real discount rate and tenor. This is a simple present value, not a debt sizing; a cover-ratio treatment would give a smaller financeable sum at the same inputs, and the finance is assumed to run from the first year of construction rather than to be drawn per tranche.

The **capture share** — the fraction of the saving accruing to the investor rather than remaining with the customer — is a lever because it moves the answer more than the discount rate does, and because leaving some saving with the customer is the proposition that wins the connection in the first place.

### 11.5.1 The deployment period

Capital is spent across a user-set build period, discounted where it falls, and the saving ramps in proportion to the fraction of the fleet complete. This replaces the earlier simplification in which the fleet existed from the first year while savings accrued in full.

The correction is worth making for its own sake rather than for its effect: across the whole 5–20 year range it moves coverage by about a point. Lengthening the build reduces the present value of the saving and the present value of the capital by nearly the same proportion, and the two effects very largely cancel. Reporting that they cancel is more useful than either number alone, because it closes the obvious objection — that the panel assumes a fleet which does not exist — without the answer depending on the assumption.

### 11.5.2 Cost improvement, and why it is not a rate

Unit capital is allowed to fall across the build. The lever is expressed as a **first-of-a-kind to mature-supply-chain level shift reached by the last year of the build**, defaulting to 20% and capped at 40%, applied linearly across the build period.

It is deliberately not a compounding annual rate. A rate of the kind sometimes proposed — 15% a year — compounds across a decade to leave unit capital at 23% of its starting value, a fall of more than three-quarters. Nothing in heat infrastructure has delivered that. Manufactured commodities with steep learning rates and orders-of-magnitude volume growth have; trench excavation and drilling have not, and offshore wind capital cost rose between 2021 and 2024. Expressing the improvement as a level shift to maturity, anchored on the Arup/DESNZ first-of-a-kind and next-of-a-kind drilling ranges [32], describes what actually happens to a supply chain and cannot be pushed to an implausible place by a long build period.

**The improvement applies to plant and subsurface only.** Trenching and building connections are excavation and labour, and do not learn. At the published default they are 36% of the build, and they are untouched by this lever at any setting. This is the mechanism behind the finding of Section 11.6: the gap does not close by waiting.

### 11.6 The published default, and what it shows

The default position — 70% network share, 70% capture, 6% real over 30 years, £250 per annual MWh distribution, £200 connections, a ten-year build and a 20% cost improvement — is the case Causeway publishes. It returns approximately **40% coverage**: of order £22bn of finance serviced against a £56bn build, leaving a gap of order £34bn.

That the figure sits well below 100% is deliberate and it is the finding. A stated gap is a policy question honestly put and survives scrutiny; a self-financing claim would not.

Two properties of the gap matter more than its size.

**It narrows with the mix, not with time.** Moving the fifth toward networks and away from individual ground-source installation improves coverage substantially; waiting does not, because the components that would have to get cheaper are excavation and labour (Section 11.5.2).

**It is not differentiated by instrument.** The figure is the arithmetic shortfall between the capitalised saving and the build cost. It says nothing about how that shortfall would be met — grant, levy, regulated return, or private capital at a return the saving alone does not support — and it is therefore not a public-expenditure estimate. Any comparison with a funding envelope requires that decomposition first, and this document does not make it.

This result aligns with the author's commercial interest, which is stated here so the reader can weigh it. The working is published for the same reason.

### 11.7 Limitations specific to this layer

The class anchors are benchmarks from jurisdictions with mature supply chains, applied to a UK deployment that does not yet exist at scale. The distribution and connection adders rest on a thin outturn sample dominated by gas-CHP-era schemes, and no robust UK outturn benchmark for *cooling* distribution was located. The network share is split evenly between ATES and doublets as a placeholder (†) pending a deployment mix. Build cost assumes the fleet exists from day one, while savings accrue only as it is built. Operations and maintenance of the new asset are excluded. And the saving itself is a price differential: it shrinks if the gas–electricity spread narrows, without a single kilowatt-hour changing.

### 11.8 A calculation considered and not published

The electrification limit of Section 10.5 implies avoided electricity generation capacity: at the binding hour, the network route's system requirement is of order 5 GW below the air-source route's for the same heat, and roughly 7.5 GW below on a gross coldest-hour basis. Priced at Capacity Market clearing prices — the published market price for de-rated firm capacity — the avoided capacity capitalises to under £1bn at the default mix, against a coverage gap an order of magnitude larger. It is additive to the bill saving rather than double-counted, since the spark gap of Section 6.2 explicitly excludes network and policy costs while this is exactly those. It was not published because a term in the rounding does not justify a further lever on an already dense panel. It is recorded here so the question is not mistaken for an oversight.

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## 12. The frontispiece (v8)

The site opens with six figures setting out the strategic case, in a deliberate order: the scale of the problem, the current cost position, the capacity feasibility of each route, the cooling corollary, the capital position, and the international comparison.

Two properties are load-bearing and are enforced by automated tests.

**Every figure is computed from the panels below it.** None is typed, and none is a separate derivation. In particular the capital figure reads the panel of Section 11 through the same function that renders it, so the summary and the detail cannot drift apart.

**The frontispiece is fixed to the trailing twelve months** regardless of the window the reader selects for the estimator below it. The two figure sets on the opening screen therefore behave differently, which is stated in a footnote beneath the panel. The alternative — letting the strategic case restate to a one-week window — would show a near-zero saving in an August week beneath a claim about the energy transition.

The capital figure is displayed in the warning colour rather than the affirmative one, because it is the figure that argues against the case. Its wording states the fraction rather than characterising it: at 40% coverage, “two fifths” is accurate where “most” would not be.

Each figure is also stated on the same basis as the figures beside it. The combustion share in the first point is computed over the same trailing twelve months as the energy, cost and emissions totals it accompanies, not over the current week — in an August week the share is materially lower, and mixing the two inside a single point would be a trap for the reader rather than for the author.

The sixth point closes with the only inference in the six: that all the technology involved is proven, and that scaling geothermal heat and cooling in the United Kingdom is therefore a diffusion challenge rather than an innovation deficit. It runs on from the international comparison rather than standing apart, because it is the conclusion that comparison forces.

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### 13. Automation, failure handling and reproducibility

A scheduled GitHub Actions workflow executes daily at 03:43 UTC: it retrieves each feed, fits the regressions, computes the balance, and commits a single JSON data file rendered by a static page. No authentication is required for any source; all are open data. The repository is public, so every computation behind every published number is inspectable. The data file carries the weekly trend history of Section 9, so the daily commit log doubles as its backup and audit trail. Each feed is wrapped so that a failure retains the last valid value and flags the affected panel; sources that publish on a lag are shown as provisional when their latest value is stale. Publication identifiers are resolved from source catalogues at runtime, and data units are inferred from magnitude, so that upstream renaming or unit changes surface as diagnostics rather than silent errors. Build failures raise an explicit notification, so that the absence of an alert is itself evidence that the pipeline is current.

**Regression suites.** Beyond the gates of Section 10.8, the build runs suites that execute the page's renderers and the estimators against a live payload rather than merely parsing them — a syntax check proves a file parses, not that a function survives its arguments. These cover the page render, the energy bars in every window, the capital panel's published default and the direction of every lever, the frontispiece and its agreement with the capital panel, the withdrawn premium diagnostic, and the weekly and hourly estimators.

**Two version counters.** A history schema version tracks new *fields* on the stored weekly series; an anchor epoch tracks a changed *basis*. A schema bump adds fields and restates stored weeks so the fields appear; an epoch bump rewrites values wherever a constant has changed what a past week would have computed. A schema bump alone would leave the series half on one basis and half on another.

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### 14. Principal limitations

The following are the material limitations, stated plainly:

- All live figures are model estimates from temperature regression on operational data calibrated to annual statistics, not metered national heat.
- **The calibration ratio currently sits outside the  $\pm 10\%$  publication gate** (Section 3.2), at approximately 1.10. It is reported openly on the site rather than rescaled, and the split is under review.
- The gas-to-heat decomposition assumes a stable demand–temperature relation; it degrades in atypical weather and under price shocks, and cannot cleanly separate cooking and non-heating gas.
- The observed and ECUK-anchored cooling estimates differ by an order of magnitude; a year-round regression diagnostic now decomposes the gap, but its weather-flat remainder cannot separate ventilation from base-load cooling (Section 4.5) (†).
- The comfort-deficit stock fractions and thermal-response coefficients carry a several-fold range (Section 5.1).
- Geothermal output is not nationally metered; the present-day bar is the most weakly constrained figure on the site (Section 8.2).



- The historical series covers only the window the live gas feed serves (~13 months); estimator constants without weekly series are applied uniformly across it (Section 9.2).
  - Annual anchors (ECUK, DUKES) lag by roughly nine to fifteen months; the live layer is provisional until each annual release.
  - Hourly heat shape is modelled from weather, not metered; the summer damping threshold and the flow-temperature conventions are estimates (†, Section 10.2–10.3).
  - The capacity line is a static overlay on one winter of observed weather, with a de-rated block that must be revised each autumn; it cannot represent storage duration or network constraints (Section 10.6–10.7).
  - The electrification limit is a peak-capacity test, not an energy test, and distribution networks are unmodelled (Section 10.5).
  - Displaced resistive heating is space-shaped; night-storage tariff profiles would reduce the credit at peaks, so gross and net figures bound the truth between them (Section 10.4).
  - The capital layer is a calculator with user-set inputs, not an estimate, and carries the specific limitations of Section 11.7. Its cost-improvement lever is a level shift to supply-chain maturity, not a forecast, and applies only to plant and subsurface; its gap is an arithmetic shortfall undifferentiated by instrument, and is not a public-expenditure estimate (Section 11.6).
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*Every author estimate on the dashboard carries a dagger (†) and an invitation. Corrections and challenges to any figure, source or method in this document are welcomed at [contact@causewaygt.com](mailto:contact@causewaygt.com).*

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